ECEN/CSEN 123 Prelab 5

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Pre-Lab Report:

• Determine the range of A/D converter integration times that can be configured on the TCS3472.
Number of integration steps is 1 to 256

• Determine the default A/D converter integration time that can be configured on the TCS3472.
Integration Time = 2.4 ms × (256 − ATIME). Reset value of ATIME is 0xFF which is 255. Thus the default integration time is 2.4ms.

• Determine the typical forward voltage drop at 20mA for the C5SMF-BJF-CR0U0351
The forward voltage drop of the blue LED at 20 mA is ~3.4V

• Determine the typical luminous intensity at 20mA for the C5SMF-BJF-CR0U0351
The typical luminous intensity of the blue LED at 20 mA is ~1.2 cd

• Determine the typical forward voltage drop at 15mA for the C4SMA-RGF-CT34QBB2
The typical forward voltage drop at 15mA (for red LED; blue and green are 10mA) is 2.0V.

• Determine the typical luminous intensity at 15mA for the C4SMA-RGF-CT34QBB2
The min luminous intensity can range from 1310-1415mcd (1.3-1.4 cd) while the max can range from 1976-2130 mcd (1.9-2.1 cd)

• Determine what color light the Vishay TEPT5700 is most sensitive to.
The wavelength of peak sensitivity is 570 nm which corresponds to yellow light.

• Determine the relative sensitivity of the Vishay TEPT5700 in response to the C5SMF-BJF-CR0U0351 as compared to the C4SMA-RGF-CT34QBB2.
The C5SMF-BJF-CR0U0351(blue led) produces a maximum wavelength of 475 nm while the C4SMA-RGF-CT34QBB2(red led) produces a maximum wavelength of 624 nm. Since the Vishay TEPT5700 has a peak sensitivity of 570nm, it will be more sensitive to the C4SMA-RGF-CT34QBB2(red led).

• Include a selfie from your planning meeting of your group members along with a rainbow colored picture from Jun 26, 2015.